

Respiratory Failure

**RESPIRATORY
FAILURE:**
CAUSES, SYMPTOMS & TREATMENT



prepared by
Dr/ Naglaa Nassar

Respiratory Failure

Outlines

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Respiratory Failure

Definition: Known as acute lung failure, is the inability of the respiratory system to maintain adequate gas exchange.

Classification of Respiratory Failure

Type I Respiratory Failure: Hypoxemic

normocapnic failure or oxygenation failure, the body fails to deliver an adequate supply of oxygen to the arterial blood. The patient presents with a low PaO₂ less than 60 mmHg and a normal PaCO₂.

Type II Respiratory Failure: Hypoxemic

hypercapnic failure or ventilatory failure, the body ineffectively eliminates carbon dioxide (CO₂) from the arterial blood. The patient presents with a low PaO₂ less than 60 mmHg and PaCO₂ is greater than 50 mmHg and pH is less than or equal 7.30.

Etiology of Respiratory Failure

- **Extra-pulmonary causes** include disorders that affect the **brain**, the **spinal cord**, the **neuromuscular system**, the thorax, the pleura, and the upper airways.
- **Intra-pulmonary causes** include disorders that affect the lower airways and **alveoli**, **the pulmonary circulation**, and **the alveolar-capillary membrane**.

Etiology of Acute Lung Failure

Extra-pulmonary

Brain

- Drug overdose (opioids, sedatives)
- Brain trauma or lesion
- Postoperative general anesthesia depression

Spinal cord

- Guillain-Barre syndrome Poliomyelitis
- Amyotrophic lateral sclerosis
- Spinal cord trauma or lesion

Neuromuscular system

- Myasthenia Gravis
- Neuromuscular-blocking agents
- Organophosphate poisoning
- Muscular dystrophy

Cont, Etiology of Acute Lung Failure

Extra pulmonary

Thorax

- Massive obesity
- Chest trauma

Pleura

- Pleural effusion
Pneumothorax
- Malignancy

Upper airways

- Trachea obstruction
- Epiglottitis
- Vocal cord paralysis

Cont, Etiology of Acute Lung Failure

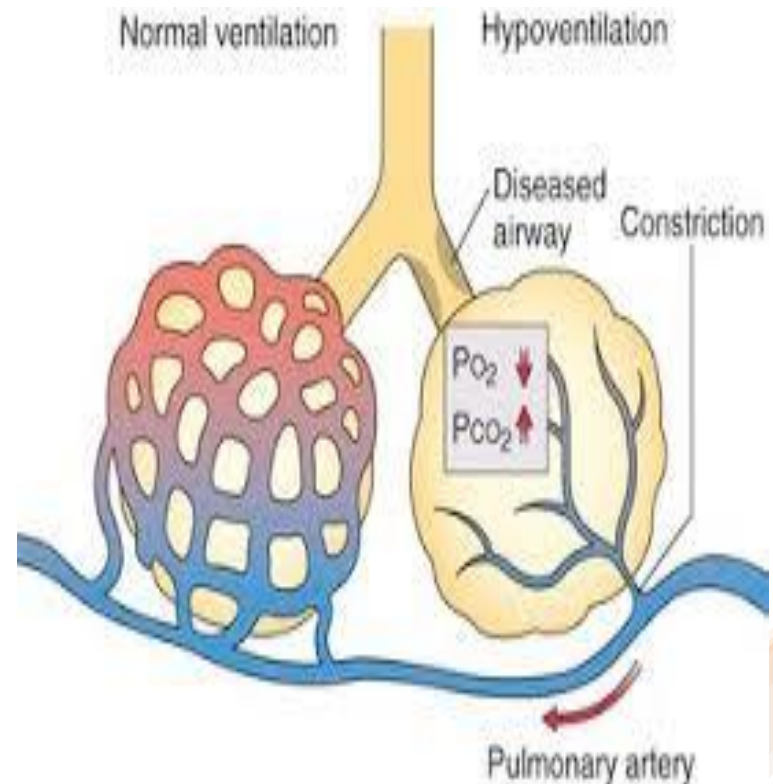
Intrapulmonary	
Lower airways and alveoli	➤ Chronic Obstructive Pulmonary Disease ➤ Asthma ➤ Bronchiolitis ➤ Cystic fibrosis ➤ Pneumonia
Pulmonary circulation	➤ Pulmonary Emboli ➤ ARDS
Alveolar-capillary membrane	➤ Inhalation of toxic gases ➤ Near-drowning

Pathophysiology of Respiratory Failure

- **Hypoxemia** is the result of impaired gas exchange and is the **hallmark** of ARF.
- The main *causes of hypoxemia* are:
- **Alveolar hypoventilation**
- **Ventilation/perfusion (V/Q) mismatching**
- **Intrapulmonary shunting**

1. Alveolar hypoventilation

It occurs when the amount of oxygen being brought into the alveoli is insufficient to meet the metabolic need of the body. This can be the result of increasing metabolic oxygen needs or decreasing ventilation. Hypoxemia caused by alveolar hypoventilation is associated with hypercapnia and commonly results from extra-pulmonary disorders.



2. Ventilation/ Perfusion (V/Q) Mismatching

- Occurs when ventilation and blood flow are mismatched in various regions of the lung in excess of what is normal. V/ Q mismatch occurs when some perfused alveoli are flooded or collapsed as occurs with pneumonia and pulmonary edema (V<Q)

- some alveoli are ventilated but not perfused because of capillary thrombosis as occurs in pulmonary embolism ($V > Q$).

3. Intrapulmonary shunting

- Shunt refer to the state in which pulmonary capillary perfusion is normal, but alveolar ventilation is lacking. Intrapulmonary shunt occurs when pulmonary capillary blood passes a nonfunctioning alveoli unit cannot pick up oxygen from the alveoli or reaches the arterial system without participating in gas exchange.

- The mixing of unoxygenated (shunted) blood and oxygenated blood lowers the average level of oxygen present in the blood. This may be the result of
- Alveolar collapse secondary to atelectasis or
- Alveolar flooding with pus, blood or fluid.

- If hypoxia progress can result in a decrease of oxygen at the cellular level. As the tissue demands for continue and the supply diminishes, oxygen supply/demand imbalance occurs and tissue hypoxia develops. Decreased oxygen to the cells contributes to impaired tissue perfusion and the development of lactic acidosis and multiple organ dysfunction syndrome.

Manifestation of Respiratory failure

- are usually related to development of hypoxemia, hypercapnia and acidosis.
1. Central Nervous System Manifestations include: Restlessness, agitation and confusion.
 2. Cardiovascular Manifestations: Tachycardia and dysrhythmias.
 3. Pulmonary Manifestations: Tachypnea, dyspnea, shortness of breath and use of accessory muscles.
 4. Renal Manifestations: Decrease urine output and edema.

Cont, Manifestation of Respiratory failure

5. *Gastro-intestinal Manifestations:* Decrease

bowel sounds, nausea and vomiting and anorexia.

6. *Skin Manifestation:* Pallor and cyanosis.

Diagnostic tests

1. **ABG** analysis conform the level of PaCO₂ of 50 mmHg or greater, the Pao₂ less than 60 mmHg and blood PH less than 7.30.
2. **Bronchoscopy** for airway surveillance or Specimen retrieval
3. **Chest radiography**
4. **Thoracic ultrasound**
5. **Thoracic computed tomography (CT)**
6. **Selected lung function studies**

Management of Acute Respiratory Failure

Aimed to:

- **Correct underlining cause**
- **Promoting adequate gas exchange**
- **Correcting acidosis**
- **Initiating nutrition support**
- **Preventing complications**

1. Oxygenation

Improve oxygenation through :

- ❖ Supplemental oxygen administration, either with a low flow system or a high flow system
- ❖ Use of positive airway pressure to correct hypoxemia within the arterial hemoglobin oxygen saturation greater than 90% without cause hypercapnia or oxygen toxicity.
- ❖ Positive pressure is necessary to open collapsed alveoli and facility their participation in gas exchange. Positive pressure is delivered by invasive or non invasive (by the mask) mechanical ventilation.

2. Ventilation

- Intervention to improve ventilation includes the use of invasive and noninvasive mechanical ventilation depending on causes and severity of disease. Patients with a PH less than 7.25 need for invasive mechanical ventilation and started on volume ventilation in the assist/ control mode.

3- Pharmacology

- Medication to facilitate dilation of the airway and may be also beneficial in the treatment of patients with ARF.
 - ❖ Bronchodilators such as B2 agonists and anticholinergic agents aid in smooth muscle relaxation.
 - ❖ Steroids administer to decrease airway inflammation and enhance the effect of B2 agonists.
 - ❖ Sedation used to assist with maintaining adequate ventilation. It can be used to comfort the patients and decrease work of breathing, if the patient is fighting the ventilator.

❖ *Analgesics* should be administered for control pain.

❖ *Neuromuscular paralysis* may be necessary to facilitate optimal ventilation and decrease oxygen consumption in the severe compromised patient.

4. Acidosis

- Acidosis may occur due to hypoxemia, and impaired ventilation causes impaired tissue perfusion which leads to the production of lactic acid and the development of metabolic acidosis. Impaired ventilation leads to accumulation of carbon dioxide and development of respiratory acidosis.

➤ To treat respiratory acidosis administer oxygen

because once the patient is adequately oxygenated and ventilated, the acidosis should correct itself.

➤ Intubation and mechanical ventilation will be

necessary for management of respiratory acidosis,

sever hypoxemia and depression of respiratory

center

- ❖ Sodium bicarbonate therapy shifts the oxygen – hemoglobin dissociation curve to the left and can worsen tissue hypoxia.
- ❖ Sodium bicarbonate may be used if the acidosis is severe (PH less than 7.10), refractory to therapy, and causing dysrhythmias or hemodynamic instability.

5- Nutrition Support

The goal of nutrition support are:

- Meet the overall nutrition needs of the patient while avoiding overfeeding.
- Avoid nutrition delivery related complication and
- Improve patient outcomes.

➤ The enteral route is preferred method of nutrition administration. If the patients cannot tolerate enteral feeding, the parenteral route will be started.

➤ Nutrition support should be initiated before the third day of MV for the well nourished patient and within 24h for the malnourished.

Complications

- ❖ **Ischemic anoxic encephalopathy**: result from hypoxemia, hypercapnia, and acidosis.
- ❖ **Dysrhythmias** are precipitated by hypoxemia, acidosis, electrolytes imbalances.
- ❖ ***Venous thromboembolism***: precipitated by venous stasis resulting from immobility and can be prevented through the use of intermittent pneumatic compression device and low dose of unfractionated heparin or low molecular weight heparin.

Cont, complication

- ❖ ***GIT bleeding*** can be prevented through the use of histamine receptor antagonists, Cytoprotective agent or proton pump inhibitors.
- ❖ ***Others complication*** associated with an artificial airways, mechanical ventilation, enteral or parenteral nutrition and peripheral arterial cannulation.

NURSING DIAGNOSES

Acute Lung Failure

- 1. Impaired Gas Exchange related to alveolar hypoventilation.**
- 2. Impaired Gas Exchange related to ventilation-perfusion mismatching or intrapulmonary shunting.**
- 3. Ineffective Breathing Pattern related to musculoskeletal fatigue or neuro-muscular impairment.**
- 4. Risk for Aspiration.**
- 5. Imbalanced Nutrition: Less Than Body Requirements related to lack of exogenous nutrients and increased metabolic demand.**
- 6. Risk for Infection.**

Optimizing Oxygenation and Ventilation through

- 1. Positioning**
- 2. Preventing desaturation**
- 3. Promoting secretion clearance.**

1. Positioning

- 1. Patients with V/Q mismatching, positioning is used to facilitate better matching of ventilation with perfusion to optimize gas exchange.**
- 2. The goal of positioning is to place the least affected area of the lungs in the most dependent position.**

3. Patient with unilateral lung disease should be positioned with the healthy lung in a dependent position.

4. Patients with diffuse lung disease should be positioned with the right lung down, because it is larger and more vascular than the left lung.

5. Patients with alveolar hypoventilation to facilitate ventilation, place in nonrecumbent position such as sitting or semisitting position.

6. Frequent repositioning every 2h is beneficial on patient's ventilatory pattern and V/Q.

2. Prevention of Desaturation

Intervention to minimizing oxygen consumption includes:

1. Limiting the patient's physical activity through

- **Performing procedures only as needed, hyperoxygenating before suction.**
- **Providing adequate rest and recovery time between procedures.**
- **Administering sedation to control anxiety**
- **Providing measure to control fever**
- **Continuous monitoring of pulse oximeter to warn of signs of desaturation**

Promoting Secretion Clearance

- ✓ Provide adequate systemic hydration and humidifying supplemental oxygen.
- ✓ Coughing and suction and postural drainage and chest percussion and vibration.
- ✓ The patient's thorax should be maintained in alignment and the head of the bed elevated 30 to 45 degree.
- ✓ This position best accommodates diaphragmatic descent and intercostals muscles action.

- ✓ **Once the patient's is extubated:** Deep breathing and incentive spirometry should be started as possible.
- ✓ **Incentive spirometry involves** having the patient's take at least 10 deep effective breaths per hour using incentive spirometry. This action help prevent atelectasis and re expand any collapsed lung tissue.
- ✓ **The chest should be auscultated** during inflation to ensure that all dependant parts of lung are well ventilated.
- ✓ **Coughing should be avoided** unless secretions are present because it promotes collapse of smaller airways.

Education of the Patient and Family

- **At discharge**, teaching should focus on the intervention necessary for preventing the reoccurrence of the precipitating disorders
- If patient's smokes, he/she should be encouraged to stop smoking and referred to a smoking cessation.
- Stress the importance of precipitating in a pulmonary rehabilitation.